Project 06 | Synthetic data | Independent, AI-assisted

Prompt-to-Presentation AI

Dynamic Executive Reporting Platform: documentation, architecture, test evidence and interview pack

Author Nipun Gupta

Live demo https://banking-ai-portfolio-bbh7cdwafuzerlfuafrmjr.streamlit.app

Source https://github.com/nipunrg1990rocks-dev/banking-ai-portfolio (projects/p6_prompt_to_presentation)

Date 31 October 2026

Fictional bank (Ledgerline Bank). All shipped data is synthetic. Not affiliated with any employer or real financial institution.

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1. Overview

One governed master dashboard replaces spreadsheet formula maintenance, chart copying and static PowerPoint rebuilding. A stakeholder types a request such as *"Create a six-slide APAC Q3 risk pack, compare Q2, rank top exception drivers, add commentary and speaker notes"*, gets a **genuinely editable .pptx** in seconds, then follows up (*"change APAC to EMEA and add high-severity exceptions"*, *"replace slide 4 with ageing"*, *"refresh using latest data"*). Every version is kept, every number is traceable to a governed query, and nothing is approved for distribution without a second person.

> Independent, AI-assisted portfolio project for a **fictional bank (Ledgerline Bank)**. All shipped data is synthetic. Not affiliated with any real institution.

What it does

Capability	Where
Two governed datasets: **operational risk** (21 months, 3 regions, 4 business lines, 5 processes, seeded anomalies) and a **technology workforce roster** with the same structure as a monthly resource tracker (the DB case-study shape)	`deckpilot/domains/`, `data/`
Validation gate: schema, category allowlists, ranges, duplicates, non-additive volume checks, formula-text neutralising, quarantine with reasons, upload limits; a new file only goes live after you activate it	`deckpilot/validation.py`, domain `load()`
Master dashboard with 5+ filters per dataset, 8 KPI tiles, trends, drivers, ageing, remediation, on-time, data quality; "create deck from this view"	`app.py`
Prompt box → validated `DeckSpec` (Pydantic + allowlists). Offline rules parser by default; Claude only if an approved key is configured, with the same validation and automatic fallback	`deckpilot/parser.py`, `deckpilot/spec.py`
Follow-ups: change scope, show only X, add/remove months, compare YoY / prior period / a named quarter, replace or regenerate slide N, add/remove slides, resize, fresh copy, refresh on latest data	`apply_followup()`
Allowlisted SQL only: parameterised DuckDB queries over approved tables and columns, each with a fingerprint written to provenance	`deckpilot/store.py`
Editable PowerPoint: native charts (incl. native waterfalls), native tables, KPI tiles, cards, footnotes, branded templates, speaker notes with talk track, evidence, caveats and provenance; DRAFT / APPROVED stamps	`deckpilot/render.py`
Templates: Executive Summary, Risk & Controls, Operations Performance (ops risk); Workforce Review, Workforce Executive Summary, Cost & Vendor (workforce)	domain `templates`
Evidence-linked commentary: each sentence carries metric, value, window and filters; interpretations labelled *Hypothesis*; missing data, quarantine and stale/planned months disclosed	domain slide builders
Versioning: append-only versions, each with prompt, parser mode, spec, dataset version + SHA-256, filters, period, query fingerprints, KPI definitions, file hash, review status	`deckpilot/versioning.py`
Review: editable commentary (saved as a new version, unverified numbers flagged), review-comment panel exported to speaker notes and a CSV review log, maker-checker approval	`deckpilot/service.py`
Hash-chained audit log of every dataset, prompt, rejection, version, edit, comment and decision	`portfolio_core/audit.py`
PDF export via LibreOffice where installed	`render.to_pdf()`

Verified results (this build, 31 Oct 2026)

Check	Result
Automated tests (`pytest projects/p6_prompt_to_presentation`)	**60/60 passed**: metrics, validation, parser and follow-ups, injection resistance, query allowlist, PPTX integrity, versioning, review controls, audit tamper detection, Streamlit AppTest (details)
Whole repository suite	194 passed, 26 skipped (Project 02 Spark tests skip without Java locally; CI runs them)
Browser test (Playwright, acceptance demo A–G against the running app)	**11/11 steps passed**; screenshots in docs/screenshots
Scripted acceptance demo (`run_demo.py`)	A–G complete in about 12 s with PDF export; output in sample_output
Reconciliation with the original case-study roster (local only, file not committed)	**13/13** published figures reproduced (134 on contract, EUR 610,393.55 Q1 T&M India, ...)

Check	Result
Live Claude parsing	Built and unit-tested with a fake client; **not run against the real API** (needs spend approval)
Hosted URL	https://banking-ai-portfolio-bbh7cdwafuzerlfuafrmjr.streamlit.app (deployed and smoke-checked by Nipun on 31 Oct 2026: generate, follow-up download, audit chain)

Acceptance demo

Step	How
A	Prompt tab → example 1 → **Generate deck** → `D-0001 v1`, six slides
B	Follow-up chip **Change APAC to EMEA and add high-severity exceptions** → `v2` (seven slides)
C	Review & versions → both versions listed, each with its own file and hash
D	Data & refresh → **Use the sample modified dataset** → validation report (9 rows quarantined, Oct 2026 added) → **Activate** → follow-up **Refresh using latest data** → `v3` on `ops-v2-…`
E	Review & versions → edit the commentary slide → **Save as new version**
F	**Download editable .pptx** (and **Export PDF**)
G	Provenance & tests → audit chain, provenance JSON, test results

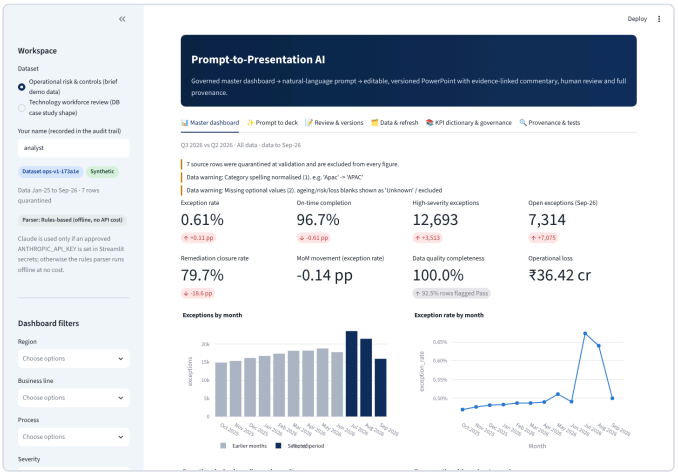
Run it

```
pip install -r requirements.txt
streamlit run projects/p6_prompt_to_presentation/app.py
pytest projects/p6_prompt_to_presentation                # tests
python projects/p6_prompt_to_presentation/run_demo.py    # scripted A–G demo (Markdown report)
node scripts/e2e_smoke_p6.js http://localhost:8501       # browser test (needs Playwright)
python projects/p6_prompt_to_presentation/generate_data.py  # regenerate synthetic data
```

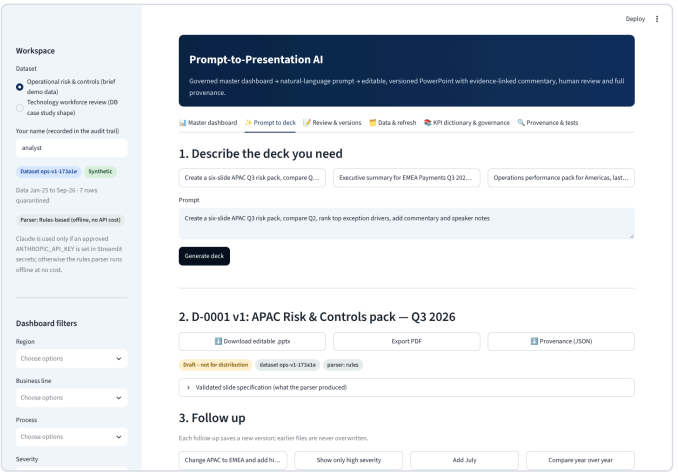
Deploy (Streamlit Community Cloud, free)

1. https://share.streamlit.io → **Create app** → repository `nipunrg1990rocks-dev/banking-ai-portfolio`, branch `main`, main file `projects/p6_prompt_to_presentation/app.py`, Python 3.12.
2. `packages.txt` in this folder asks Streamlit Cloud to install LibreOffice for PDF export. If the build is slow or fails, delete that file: the app still works and the PDF button explains that PDF is unavailable on that host.
3. No secrets are needed. Live Claude parsing is off until an `ANTHROPIC_API_KEY` is added under app **Settings → Secrets** after spend approval.

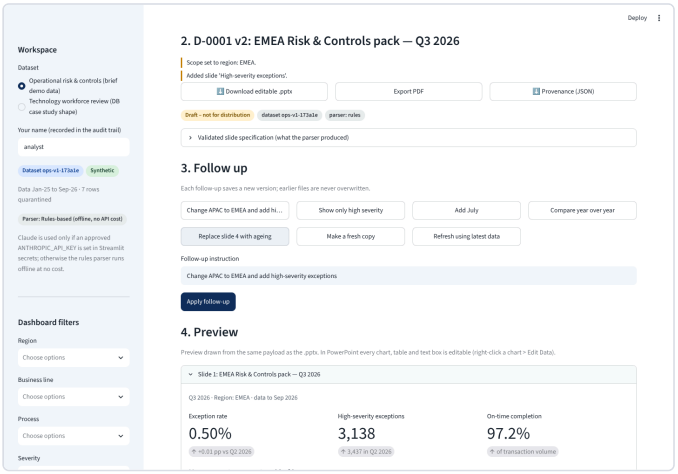
2. Screens



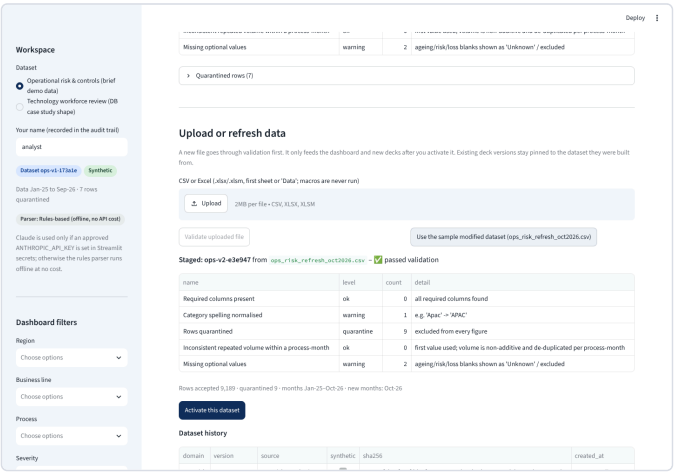
Master dashboard (ops risk)



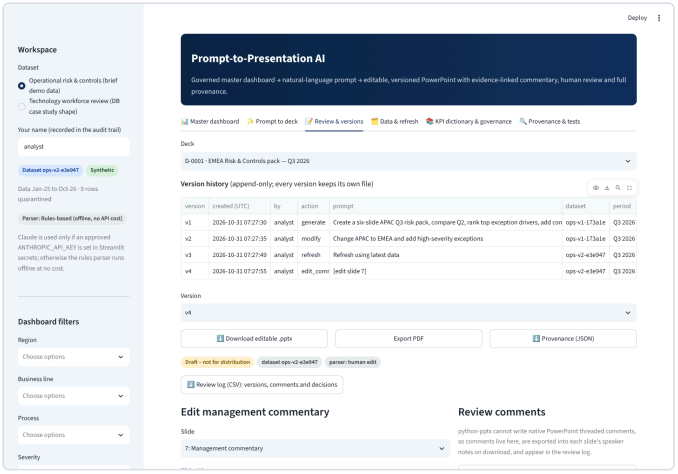
A: APAC Q3 six-slide deck



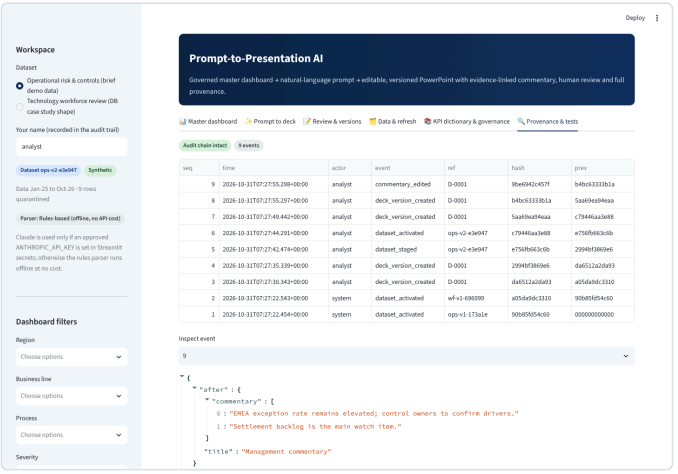
B: follow-up to EMEA + high severity



D: modified dataset validation



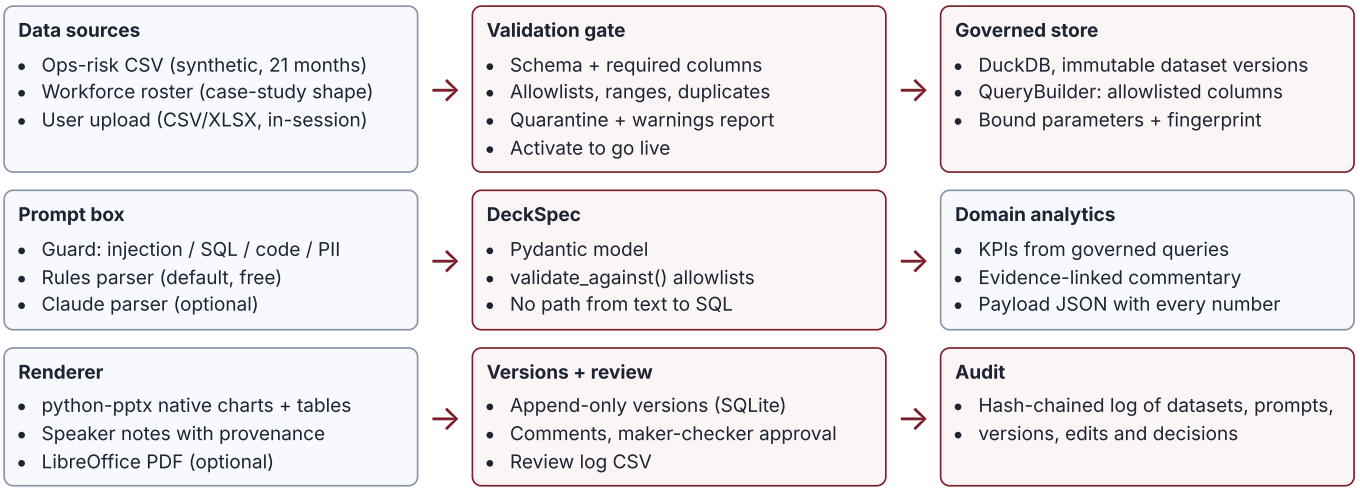
E: commentary edited, new version



G: provenance and audit chain

3. Architecture and data flow

Component view



Dark red boxes are controls.

End-to-end data flow for one prompt

1. **Stakeholder** types: *six-slide APAC Q3 risk pack, compare Q2, rank drivers, commentary*
2. **Guard** checks length, injection phrases, SQL/code and PII; rejections are logged
3. **Parser** (rules, or Claude if approved) proposes template, filters, months, comparison, slides
4. **DeckSpec validator** accepts only allowlisted values that exist in the active dataset
5. **QueryBuilder** runs parameterised SELECTs over approved columns and records fingerprints
6. **Domain** computes KPIs and writes commentary, each sentence linked to its evidence
7. **Renderer** writes a new .pptx (never overwrites) and hashes it; version D-0001 v1, DRAFT
8. **Follow-up** *change APAC to EMEA and add high-severity exceptions* becomes v2 on the pinned dataset; v1 is kept

Key design decisions

Decision	Why
Prompt → **validated spec**, never prompt → SQL	The model (or rules) can only choose from allowlisted templates, slide types, dimensions and values. There is no path from user text to an executed query, so prompt injection cannot reach data.
Rules parser by default, Claude optional	Works with zero cost and no credentials; Claude improves phrasing coverage when an approved key exists, and its JSON goes through the same validator with automatic fallback.
Payload JSON between analytics and rendering	Every number on a slide is stored with the version, so approval re-renders the identical content with an "APPROVED" stamp, and provenance can be audited without re-querying.
Native charts and tables	Charts are editable in PowerPoint (Edit Data). Waterfalls are stacked columns with an invisible base series; a Sankey and Marimekko from the original case-study deck are replaced by stacked bars so they stay editable.
Versions are **append-only** and **pinned to a dataset**	A follow-up keeps the parent's dataset unless the user says "refresh using latest data", so numbers never change silently; every version keeps its own file and hash.
Volume is **non-additive**	In the ops-risk data, transaction volume repeats on every cluster row; KPIs de-duplicate at process-month before summing, and severity filters narrow only the numerator.
Review comments in a **panel**, exported to notes + CSV log	python-pptx cannot write native threaded comments; the platform says so and exports comments transparently.
Maker-checker approval	The author cannot approve, open comments block approval, and edited commentary containing numbers not found in the slide data must be confirmed as verified.

Where things live

Path	Purpose
`app.py`	Streamlit UI (6 tabs)
`deckpilot/spec.py`	Pydantic `DeckSpec`, period labels, comparison windows
`deckpilot/parser.py`	Guard, rules parser, follow-up interpreter, optional Claude parser
`deckpilot/validation.py`	Upload reading, limits, validation report, quarantine
`deckpilot/store.py`	Dataset versions, DuckDB query builder with allowlists and fingerprints
`deckpilot/domains/ops_risk.py`	Ops-risk data contract, KPIs, slides, synthetic generator
`deckpilot/domains/workforce.py`	Workforce roster loader, KPIs, case-study slide set, synthetic roster
`deckpilot/payload.py`	Slide payload contract, formatting, safe division, unverified-number check
`deckpilot/render.py`	python-pptx renderer and LibreOffice PDF export
`deckpilot/versioning.py`	SQLite versions, comments, reviews, review log
`deckpilot/service.py`	Orchestration used by the app, tests and `run_demo.py`
`portfolio_core/` (repo root)	Shared guardrails, hash-chained audit log, Claude client

Production path (not built; needs an authorised workspace)

- Replace DuckDB with Databricks SQL over Delta tables governed by Unity Catalog (row filters for region entitlements); keep the same `QueryBuilder` contract.
- Store `.pptx` files in object storage with retention, and versions/audit in PostgreSQL.
- SSO identity instead of a typed name; reviewer roles from an entitlement system.
- LibreOffice in a sidecar container for PDF export.

Known limitations

- State is per browser session and ephemeral on Streamlit Community Cloud (a restart resets it); this is deliberate for a public demo.
- The rules parser covers the documented vocabulary; unusual phrasing is reported as "could not interpret" rather than guessed.
- Commentary is template-generated from computed metrics; it does not explain causes. Root causes are what control owners recorded, and interpretations are labelled as hypotheses.
- PDF export needs LibreOffice on the host (`packages.txt` requests it on Streamlit Cloud).

4. Data dictionary

All shipped data is **synthetic** and generated by `generate_data.py` with fixed seeds. "Ledgerline Bank" is fictional.

1. Operational risk dataset (`data/ops_risk_synthetic.csv`)

Grain: one row per `reporting_month × region × business_line × process × severity × root_cause` cluster. Jan 2025 – Sep 2026 (21 months), 3 regions × 4 business lines × 5 processes, about 8,750 rows.

Column	Type	Required	Allowed values / range	Meaning
record_id	text	yes	unique	Cluster record id
reporting_month	YYYY-MM	yes	valid month	Month-end period
region	category	yes	APAC, EMEA, Americas	Booking region
business_line	category	yes	Retail Banking, Corporate Banking, Wealth Management, Markets Operations	Owner
process	category	yes	Payments, Trade Settlement, KYC Onboarding, Reconciliations, Loan Servicing	Process
transaction_volume	int	yes	0 – 5,000,000	Process-month volume, **repeated on every row of the process-month (non-additive)**
completed_on_time	int	yes	0 – volume	Process-month count within SLA (repeated, non-additive)
exceptions	int	yes	0 – 100,000	Exceptions in this cluster
severity	category	yes	Low, Medium, High, Critical	Severity
root_cause	category	yes	8 values (Manual keying error, Missing documentation, System outage, Static data mismatch, Late counterparty confirmation, Screening false positive, Process design gap, Vendor delay)	As recorded by the control owner
ageing_days	int	no	0 – 730	Age at month end (days to close if closed)
control_owner	text	yes		Accountable role (no personal names)
risk_score	float	no	0 – 100	Residual risk score
loss_amount_inr	float	no	0 – 5bn	Operational loss, INR
remediation_status	category	yes	Open, In Progress, Closed	
data_quality_status	category	yes	Pass, Warning, Fail	Source-system DQ flag

Seeded anomalies (for testing and the demo story)

Anomaly	Where	Expected handling
Exception spike, High/Critical uplift, "System outage" driver	APAC Payments, Jul–Aug 2026	Visible on dashboard and in Q3 vs Q2 commentary
Reconciliations deterioration, "Static data mismatch"	APAC, Q3 2026	Appears in driver ranking
Ageing deterioration	Americas KYC Onboarding from Jun 2026	90+ day band grows
Large Critical loss (₹4.85 cr)	EMEA Markets Operations Trade Settlement, Sep 2026	Appears on loss and high-severity slides
3 exact duplicate rows	random	Quarantined
2 missing `ageing_days`	random	Kept, disclosed as warning / "Unknown" band
`completed_on_time` > volume	1 row	Quarantined
`region = "Apac"`	1 row	Normalised to APAC, disclosed
`region = "LATAM"`	1 row	Quarantined (unexpected category)

Anomaly	Where	Expected handling
Negative loss, risk score 140	1 row each	Quarantined (out of range)

`data/ops_risk_refresh_oct2026.csv` is the "modified dataset" for demo step D: it adds October 2026, restates APAC Payments Low-severity exceptions for September (+25%), and plants two invalid rows (month `2026-13`, severity `Catastrophic`) that the gate quarantines.

Ops-risk KPIs

metric_id	Definition	Aggregation rule
exception_rate	sum(exceptions) / sum(transaction_volume de-duplicated per process-month)	Ratio of sums; n/a when volume = 0; severity filter narrows the numerator only
on_time_pct	sum(completed_on_time) / sum(transaction_volume), both de-duplicated	Ratio of sums; unaffected by severity filter
high_severity_exceptions	exceptions where severity in High, Critical	Additive
open_exceptions	exceptions not Closed, in the latest month of the period	Point in time
ageing_band	open exceptions by 0-30 / 31-60 / 61-90 / 90+ days	Unknown band when blank
closure_rate	Closed exceptions / all exceptions in period	Ratio of sums
mom_movement	exception rate latest month − prior month	Percentage points
dq_completeness	non-blank optional cells / all optional cells; plus % rows flagged Pass	Ratio of sums
loss_inr	sum(loss_amount_inr)	Blanks excluded and disclosed

2. Workforce roster (`data/workforce_roster_synthetic.csv`)

Same 33 source columns as the case-study resource tracker: `Sr No, Last Updated By, Team, Project, Category, Position #, Name, Gender, Country, City, Email-ID, Start Date, End Date, Functional Manager, Vendor, PO, Profession Category, Profession, Profession Role, Cost Center, Jan-18 … Dec-18, Clarity Band`. Monthly columns are FTE allocations (0–1). Names, vendors and position numbers are invented.

Curated tables after validation (names and e-mails are dropped – data minimisation; a person is a hashed `person_key` of Position # + Name):

- `people`: one row per unique person-contract with location, resource type, vendor, profession, dates, daily rate, total FTE, zero-allocation flag.
- `allocation`: one row per person-month with FTE and cost (FTE × 22 × daily rate).

Seeded features: two roll-off waves (end March, end September), India-heavy reduction, 20 contracts with zero allocation all year, one duplicate record, e-mails with spaces, untrimmed project / PO / manager text, non-standard profession category spellings.

Rate card where no daily rate is given: Band_11 = EUR 200, Band_9 = EUR 100 (as in the case study).

Workforce KPIs

metric_id	Definition
people_on_contract	Unique Position # + Name with Start ≤ as-of date ≤ End; as-of = min(period end, roster date)
people_allocated	People with allocation > 0 in the month
fte	Sum of monthly allocations
monthly_cost_eur	FTE × 22 working days × daily rate (EUR); no FX, tax or mark-up
tm_share	T&M people on contract / people on contract
women_share	People on contract with Gender F / people on contract
avg_daily_rate	FTE-weighted average daily rate in the month
vendor_concentration	External (T&M + Fixed Price) people on contract from the two largest vendors / all external
retention	Share of the base-month team still allocated each month

metric_id	Definition
contracts_ending	People on contract grouped by end date
zero_allocation	Contracts whose monthly allocations are all zero

The original case-study roster is **not** in the repository. `reconcile_case_study.py` reproduces 13/13 of the published deck figures from it when run locally.

5. Test results

Last local run: 31 Oct 2026 (Python 3.13, Linux; CI re-runs on every push with Python 3.12).

Test file	What it proves	Result
test_p6_metrics.py	KPI accuracy vs an independent pandas calculation, volume de-duplication (non-additive), severity filter narrows numerator only, divide-by-zero, comparison windows, missing-period disclosure, workforce on-contract and Q1 T&M cost formula	11/11 passed
test_p6_validation.py	Seeded anomalies quarantined (duplicates, LATAM, on-time > volume, negative loss, risk 140), schema errors block activation, formula text neutralised, upload size/type limits, refresh only after activation, roster checks, xlsx round trip	7/7 passed
test_p6_parser.py	Acceptance prompt A, prompt variants, all seven follow-up types, 7 injection/SQL/code prompts rejected, unknown values never reach the spec, LLM output validated and fallback to rules	25/25 passed
test_p6_query.py	Only allowlisted tables/columns; filter values bound as parameters; stable query fingerprints	3/3 passed
test_p6_pptx.py	Decks open in python-pptx, native charts and tables (no images), notes carry provenance, every slide type renders, files never overwritten, PDF export	5/5 passed
test_p6_workflow.py	Acceptance demo A–G end to end, slide replacement keeps other slides identical, fresh copy, unverified-number flag, maker-checker / open-comment / approval rules, audit tamper detection, rejected prompts logged	7/7 passed
test_p6_app.py	Streamlit AppTest: dashboard renders, generate + follow-up, workforce dashboard	2/2 passed
Total (Project 06)		**60/60 passed**

Whole repository suite at the same time: 194 passed, 26 skipped (the skips are the Project 02 Spark tests, which need Java and PySpark locally; CI runs them).

Lint: `ruff check .` – all checks passed.

Reconciliation against the original case-study roster (run locally with `reconcile_case_study.py`, file not committed): 13/13 figures match the published deck (134 on contract, 219 people, FTE 195 → 85, 105 contracts ending 31 Dec, 67 T&M, 87 external, 54 from the top two vendors, 20 zero-allocation, Q1 T&M India EUR 610,393.55 for 72 contractors, run-rate EUR 691k → 374k).

Project 06 acceptance demo (A-G)

Step	What	Version	Slides	Dataset	Period	Filters	Note
A	Generate APAC Q3 six-slide deck	D-0001 v1	6	ops-v1-173a1e	Q3 2026	region=['APAC']	
B	Follow-up: change APAC to EMEA, add high-severity	D-0001 v2	7	ops-v1-173a1e	Q3 2026	region=['EMEA']	
C	Second file saved, first kept	D-0001 v2	7	ops-v1-173a1e	Q3 2026	region=['EMEA']	v1 file still present: True
D	Upload modified dataset, validate, activate, refresh	D-0001 v3	7	ops-v2-e3e947	Q3 2026	region=['EMEA']	9,189 rows accepted, 9 quarantined
D	Follow-up: add October	D-0001 v4	7	ops-v2-e3e947	Jul–Oct 2026	region=['EMEA']	

Step	What	Version	Slides	Dataset	Period	Filters	Note
	(now in data)						
E	Edit management commentary on slide 7	D-0001 v5	7	ops-v2-e3e947	Jul–Oct 2026	region=['EMEA']	human-edited, evidence kept for unchanged lines
F	Approved, downloaded editable .pptx	D-0001 v5	7	ops-v2-e3e947	Jul–Oct 2026	region=['EMEA']	D-0001_v05_EMEA_Risk___Controls_pack___Jul_Oct_2026_APPROVED.pptx (84,336 bytes)
WF	Workforce review (case-study replica)	D-0002 v1	11	wf-v1-696999	FY 2018	all	
WF	Follow-up: Pune T&M only	D-0002 v2	11	wf-v1-696999	FY 2018	location=['Pune'], category=['T&M']	

(G) Audit events: 18, hash chain intact: True. PDF export: yes. Total time 11.6s.

6. Interview pack

Everything here describes what was actually built and tested. Numbers come from `docs/test_results.md` and `run_demo.py`. If you change the code, re-run them before quoting.

Honest framing. This is an independent, AI-assisted portfolio project. I owned the problem definition, banking requirements, data contracts, KPI definitions, control design, acceptance criteria and validation of outputs. Much of the code was written with an AI coding assistant under my direction and review. It runs on synthetic data for a fictional bank. It is not a production system and has not been used with any employer's data.

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1. Three-minute demo script

Time	Do	Say
0:00	Open the **Dashboard** tab (ops risk).	"Executive reporting at most banks is a monthly rebuild: extract, fix formulas, copy charts into PowerPoint, rewrite commentary. This replaces that with one governed dashboard and a prompt box that produces an editable deck."
0:25	Change region to APAC, months to Q3. Point at the exception-rate trend.	"Every tile is a governed KPI. Volume in this data is non-additive, so it's de-duplicated before any ratio. The APAC Payments spike in July and August is a seeded anomaly I use to test the story."
0:50	**Prompt to deck** → example 1 → **Generate deck**.	"The prompt never becomes SQL. It becomes a validated spec: template, filters, months, comparison, slides. Anything outside the allowlist is rejected."
1:15	Show the preview and the spec. Click the chip **Change APAC to EMEA and add high-severity exceptions**.	"Follow-ups modify the spec, not the slides. Version 2 is a new file. Version 1 is untouched."
1:45	**Review & versions** → download v2 and open it in PowerPoint. Click into a chart and choose Edit Data.	"These are native charts and tables, so the team can still edit them. The speaker notes carry the talk track, the evidence for every sentence, and provenance: dataset version, hash, filters and query fingerprints."
2:15	Try to approve as the author.	"Maker-checker: the author can't approve, open comments block approval, and edited commentary with numbers that aren't in the data must be confirmed as verified."
2:40	**Provenance & tests** → Verify chain.	"Every prompt, version, edit and decision is in a hash-chained audit log. It has 60 automated tests and an 11-step browser test, and it reproduces 13 of 13 published figures from a real case-study deck."

2. Ten-minute demo script

1. Problem and goal (1 min). Monthly packs take days and break easily. They depend on formula maintenance and static decks, and nobody can say where a number came from. The goal is a single source of truth, prompt-driven decks, and governance built in, not added afterwards.

2. Data and validation (1.5 min). Go to **Data & refresh**. Show the two datasets. Ops risk covers 21 months, 3 regions, 4 business lines and 5 processes. Workforce has the same shape as a monthly resource tracker. Show the validation report: required columns, category allowlists, ranges and duplicates. "Apac" is normalised to APAC, LATAM is quarantined, completed-on-time greater than volume is quarantined, and formula text is neutralised. A new file goes live only after it is activated.

3. Dashboard (1 min). Show the filters for region, business line, process, severity and months. Cover trend, drivers, ageing bands, remediation, on-time and data quality. Then **Create deck from this view**.

4. Prompt to deck, acceptance A (1.5 min). Generate "six-slide APAC Q3 risk pack, compare with Q2, rank top exception drivers, add commentary and speaker notes". Show the parsed spec and the parser mode (rules). Then open the .pptx and walk through it. The cover has a DRAFT stamp. The KPI table carries RAG and Better/Worse labels. Then the Q3 vs Q2 comparison, drivers, the commentary slide labelled *draft*, and the appendix with KPI definitions.

5. Follow-ups, acceptance B and C (1.5 min). - "Change APAC to EMEA and add high-severity exceptions" produces v2 with seven slides. The ₹4.85 cr Critical loss in EMEA Markets now appears. - "Replace slide 4 with ageing" regenerates only that slide. - "Make a fresh copy" creates a new deck id. - Show that v1's file and hash are still there.

6. Refresh, acceptance D (1 min). Upload the sample modified dataset. The report shows Oct 2026 added, September restated and the bad rows quarantined. Activate it, then "Refresh using latest data" gives v3 on `ops-v2-…`. Explain that earlier

versions stay pinned to the dataset they were built on, so numbers never change silently.

7. Review, acceptance E and F (1 min). Edit the commentary and save it as a new version. Add a review comment, then log in as another reviewer: approval is blocked until the comment is resolved. Then approve. The approved file carries an APPROVED stamp. Download it with comments in the notes, plus the review-log CSV and the PDF.

8. Workforce domain (0.5 min). Switch dataset and generate the Workforce Review. It replicates the case-study deck: headcount bridge, location trend, cost bridge, contract cliff and vendor concentration.

9. Provenance and tests, acceptance G (0.5 min). Show the audit table, verify the chain and show the provenance JSON. Point to the test results: 60/60 tests and the CI step that runs the A–G demo.

10. Limits and production path (0.5 min). Data is synthetic. Live Claude parsing is built but has not been run against the API. For production, you would use Databricks SQL with Unity Catalog row filters, SSO, object storage, and PostgreSQL for versions.

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3. Thirty interview questions and answers

Product and business

1. **What problem does it solve?** Executive packs are rebuilt by hand every month. It's slow, error-prone and hard to audit. This turns the pack into a governed product: one dataset, one KPI dictionary, prompts to produce decks, and a full trail.
2. **Who are the users?** The Head of Operations and the CRO's office consume the decks. Analysts and PMO create them. Control owners and reviewers sign off. The maker-checker split reflects those roles.
3. **How would you measure success?** Time from data cut-off to approved pack (target: hours, not days). The number of manual corrections found in review. The share of numbers traceable to a query, which should be 100%. Reuse of templates across regions.
4. **Why PowerPoint and not just a dashboard?** Committees still consume decks, annotate them and file them as records. So the output is genuinely editable, with native charts and tables, rather than images.
5. **What did you deliberately leave out?** Free-text Q&A over data, auto-sending to distribution lists, and native PowerPoint comments, which python-pptx can't write. The comments go to the notes and a CSV review log instead, and the app says so.
6. **How did you prioritise?** Correctness and governance first: validation, allowlists, versioning and review. Then the acceptance demo. Then polish. A pretty deck with an untraceable number fails in a bank.

Data and KPIs

1. **What is non-additive volume and why does it matter?** In the ops data, transaction volume is recorded once per process-month but repeated on every exception cluster row. Summing it naively would inflate the denominator many times over and understate the exception rate. I de-duplicate at the process-month grain before summing.
2. **How does a severity filter affect the exception rate?** It narrows only the numerator. "High-severity exception rate" is high-severity exceptions over all volume, because volume doesn't have a severity.
3. **How do you handle bad data?** Every row is checked against required columns, category allowlists, ranges, date validity, logical rules (on-time ≤ volume) and duplicates. Fixable issues like "Apac" are normalised and disclosed. Others are quarantined with a reason. Warnings and quarantine counts appear on the data-quality slide and in the notes.
4. **What happens when the comparison period has no data?** The comparison is shown as n/a and the gap is disclosed on the slide and in the notes, not silently dropped. Division by zero returns n/a, not zero.
5. **How did you validate the workforce numbers?** I reconciled them against a published case-study deck. A local script compares 13 headline figures (for example, 134 people on contract and €610,393.55 Q1 T&M spend in India) with the engine's output from the original file. All 13 matched. The original file is not in the public repo.
6. **How are KPIs defined?** In a KPI dictionary that ships with the data contract. It shows the metric id, definition, aggregation rule and caveats. The dictionary is shown in the app and printed in each deck's appendix.

AI and LLM design

1. **Where does the LLM sit?** Only in parsing the prompt into a structured spec. It never writes SQL, never sees the data and never writes the numbers.
2. **Why a rules parser by default?** It's free, deterministic and testable, and works without any credentials. The Claude parser improves coverage of unusual phrasing when an approved key exists. Its JSON goes through the same Pydantic model and allowlist checks, and it falls back to rules on any failure.
3. **How is commentary grounded?** Every sentence is generated from computed metrics and stored with its evidence: metric, value, window and filters. Causal statements are labelled "Hypothesis" and cite the root causes recorded by control owners.
4. **How do you stop prompt injection?** There are three layers. A guard rejects injection phrases, SQL and code patterns. The output space is a closed schema: templates, slide types and filter values must be in allowlists and exist in the dataset. There is no path from text to an executed query.
5. **What if a user edits commentary and types a wrong number?** Numbers in the edited text are checked against the slide's data. Unmatched numbers are flagged, and approval requires the reviewer to note that they were verified.
6. **How would you evaluate the LLM parser?** With a labelled set of prompts and their expected specs, scored on exact match per field, the rejection rate on adversarial prompts and the fallback rate. I'd also track cost and latency per prompt. The rules parser tests already provide the harness.
7. **Did you use the live Claude API?** No. The client is built and tested with a fake. Running it live needs an API key and spend approval, which I haven't given.

Engineering

1. **Why a payload JSON between analytics and rendering?** It freezes every number with the version. So approval re-renders identical content with a new stamp, provenance doesn't need a re-query, and the renderer is testable on its own.
2. **How are follow-ups implemented?** As spec transformations: change filters, add or remove months, change comparison, replace or add slides. Unchanged slides are copied from the parent payload, and only the affected slides are regenerated.
3. **How is versioning designed?** Append-only in SQLite. Each version stores its prompt, parser mode, spec, dataset version and SHA-256, query fingerprints, file path and hash, and status. The renderer refuses to overwrite an existing file.
4. **Why pin follow-ups to the parent's dataset?** So the same request never silently produces different numbers after a data refresh. Refresh is an explicit action that creates a new version on the new dataset.
5. **How do you make charts editable?** They use python-pptx native chart objects with embedded data. Waterfalls are stacked columns with an invisible base series. The Sankey and Marimekko from the original deck were replaced with stacked bars so they stay editable.
6. **What does the audit log protect against?** Silent changes. Each event includes the hash of the previous one, so editing or deleting a past event breaks the chain. A test proves that tampering is detected.
7. **How is it tested?** 60 pytest tests cover metrics, validation, the parser and follow-ups, injection, the query allowlist, PPTX integrity, versioning, review rules, audit tampering and a Streamlit AppTest. There's also an 11-step Playwright browser run of demo A–G, and CI runs the scripted acceptance demo on every push.

Risk, controls and production

1. **What controls would a bank's model risk or ops risk team ask about?** Data lineage (dataset hash plus query fingerprints), access control (allowlists, which need SSO and entitlements in production), change control (versioning), maker-checker approval, audit trail and AI use disclosure. Each one is either built here or listed as a production gap.
2. **What would change for production?** Databricks SQL over Delta tables with Unity Catalog row filters for regional entitlements, keeping the same QueryBuilder contract. SSO identities and reviewer roles from the entitlement system. Object storage with retention for decks. PostgreSQL for versions and audit. LibreOffice in a sidecar for PDF. A model-risk review of the LLM parser.
3. **What about data privacy?** Only synthetic data is shipped. On the workforce roster, names and e-mails are dropped at load, and people are identified by a hash. Uploads stay in memory for the session. Prompts pass through PII redaction

before any model call.

4. **What would you do next?** Evaluate the live LLM parser on a labelled prompt set. Add role-based entitlements. Add scheduled refresh with data-freshness SLAs. Build a template designer so business users can add slide types without code.

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4. Resume and LinkedIn bullets (ATS-ready)

Project title line Prompt-to-Presentation AI: Dynamic Executive Reporting Platform (independent AI-assisted portfolio project, 2026) | Python, Streamlit, python-pptx, DuckDB, Pydantic, Claude API (optional), pytest, GitHub Actions

Resume bullets (choose 3–5) - Designed and delivered, as product owner using AI-assisted development, a prompt-to-PowerPoint reporting platform that turns natural-language requests into editable executive decks with native charts, KPI tables, commentary and speaker notes, on synthetic banking operational-risk and workforce data. - Defined data contracts, a KPI dictionary and validation rules for 21 months of synthetic operational-risk data, including de-duplication of non-additive volume, category allowlists, range checks and quarantine reporting. - Specified a governed AI architecture in which prompts map to a validated Pydantic spec rather than SQL, with allowlisted parameterised queries, prompt-injection guards and an offline rules fallback for the optional LLM parser. - Built in banking controls: append-only deck versioning, dataset pinning, maker-checker approval, unverified-number checks on edited commentary and a hash-chained audit log, with provenance recorded for each deck. - Reconciled the analytics engine against a published workforce case-study deck, reproducing 13 of 13 headline figures. Defined acceptance tests covered by 60 automated tests, an 11-step browser test and a CI-run acceptance demo.

LinkedIn "Projects" entry

> **Prompt-to-Presentation AI (independent, AI-assisted)** A governed reporting platform: one dashboard plus a prompt box that produces editable PowerPoint packs, with follow-ups such as "change APAC to EMEA" and "refresh using latest data". I owned the requirements, data and KPI design, controls and validation, and used an AI coding assistant for much of the implementation. Synthetic data, fictional bank. Features: native charts, evidence-linked commentary, version history, maker-checker review, a hash-chained audit trail, and 60 automated tests. Repo: github.com/nipunrg1990rocks-dev/banking-ai-portfolio

Do not claim: production use, real bank data, live LLM accuracy figures, user adoption, time savings measured in a real team, or that you wrote all the code by hand.